Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

What is the equivalent weight of $K_4\left[Fe(CN)_6\right]$ as per the reaction ?
$\left[Fe(CN)_6\right]^{-4}+MnO_4^- \rightarrow Fe^{+3}+CO_2+NO_3^-+Mn^{+2}$

(A) $\frac{M}{5}$

(B) $\frac{M}{61}$

(C) $\frac{\mathrm{M}}{66}$

(D) $\frac{\mathrm{M}}{36}$

Q.2

An amount of 0.014 mole of a solid is dissolved in water, 0.1 mole of KI and a few drop of dilute H_2SO_4 are added. The deep brown solution formed is titrated with a $0.1 \mathrm{M} Na_2\ S_2O_3$ solution until solution is completely discoloured, the volume of $Na_2\ S_2O_3$ solution required is $1.12 \ell t$. If the solid is any one of the following options, identify it?

(A) $NaIO_4$

(B) $NaIO_3$

(C) $NaOI$

(D) NaI

Q.3

In which of the following changes, there is a transfer of five electrons?

(A) $MnO_4^- \rightarrow Mn^{+2}$

(B) $CrO_4^{2-} \rightarrow Cr^{3+}$

(C) $MnO_4^- \rightarrow MnO_2$

(D) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

Q.4

$2.08\ \mathrm{kg}$ of an equimolar mixture of ferrous oxalate $(\mathrm{MW}=144)$ and ferric oxalate $(\mathrm{MW}=376)$ require X moles of $K_2Cr_2O_7$ in acidic medium for complete oxidation then X is -

(A) $16/3$

(B) 6

(C) $32/3$

(D) $10/3$

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Q.5

How many moles of acidic potassium permanganate reacts with one mole of $KHC_2O_4 \cdot HCOONa \cdot 2H_2O$?

(A) $\frac{2}{5}$

(B) $\frac{4}{5}$

(C) $\frac{5}{4}$

(D) $\frac{5}{2}$

Q.6

In the reaction $Al + Fe_3O_4 \rightarrow Al_2O_3 + Fe-$
what is the total no. of electrons transferred during the change-

(A) 16

(B) 24

(C) 8

(D) 12

Q.7

In the redox reaction -
$10FeC_2O_4 + xKMnO_4 + 24H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 20CO_2 + yMnSO_4 + 3\,K_2SO_4 + 24H_2O.$
The values of x and y are respectively -

(A) 6,3

(B) 3,6

(C) 3,3

(D) 6,6

Q.8

A solution containing 2.68×10^{-3} mol of A^{+n} ions requires 1.61×10^{-3} mole of $MnO^{-}{}_4$ for the oxidation of A^{+n} to $AO^{-}{}_3$ in acidic medium. What is the value of $n-$

(A) 1

(B) 2

(C) 3

(D) 4

Q.9

Which of the following is/are a redox reaction-

(A) $\frac{1}{2}H_2 + \frac{1}{2}I_2 \Leftrightarrow HI$

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(B) $PCl_5 \Leftrightarrow PCl_3 + Cl_2$

(C) $2CuSO_4 + 4KI \Leftrightarrow Cu_2I_2 + 2\,K_2SO_4 + I_2$

(D) $CaOCl_2 \Leftrightarrow Ca^{+2} + OCl^- + Cl^-$

Q.10

Choose the correct options

(A) The oxidation number of S in SO_4^{2-} ion is $+6$

(B) The oxidation number of S in HSO_3- ion is $+4$

(C) The oxidation number of Pt in $(PtCl_6)^{2-}$ is $+4$

(D) The oxidation number of Mn in $(MnO_4)^-$ ion is $+7$

Q.11

12 g of impure cyanogen undergo hydrolysis by two different pathways :
(i) $(CN)_2 + 4H_2O \rightarrow (NH_4)_2\,C_2O_4$
(ii) $(CN)_2 + 2H_2O \rightarrow NH_2CONH_2$
The exact amount of urea was obtained when 11.52 g of pure ammonium carbonate was heated. If 20ml of 1.6M acidic $KMnO_4$ solution was required to completely oxidize $(NH_4)_2\,C_2O_4$, Choose the correct options -

(A) % purity of cyanogen is 86.67%

(B) % progress in case (i) is 40%

(C) % progress in case (ii) is 60%

(D) Carbon is oxidized in case (ii)

Q.12

Paragraph:
Several features of sulfuric acid are given below.
Preparation of Sulfuric Acid: Sulfuric acid is commonly prepared by the combustion of elemental sulfur to sulfur dioxide, followed by the catalytic oxidation of sulfur dioxide to sulfur trioxide. Sulfur trioxide is then absorbed into a 98% aqueous solution of H_2SO_4, and water is added to maintain a 98% concentration. SO_3 reacts with the water in the aqueous solution according to Reaction 1. $SO_3(\,g) + H_2O(\ell) = H_2SO_4(\ell)$
Properties: Concentrated sulfuric acid is $98\%H_2SO_4$ and 2% water by mass. It has a density of 1.84 g/mL and a boiling point of $338^\circ C$.
Preparation of other acids : $HCl(g)$ and $HNO_3(\ell)$ may be prepared by the reaction between sulfuric acid and the sodium salt of the corresponding conjugate base $(Cl^-$ or NO_3^-, respectively).
Formation of SO_2 : Sulfuric acid forms SO_2 gas when it reacts with several compounds. For example, I_2 and SO_2 are formed when I^-reacts with concentrated H_2SO_4; Br_2 and SO_2 are formed when Br^-reacts with concentrated H_2SO_4.$Cu^+$and SO_2 are formed in hot solutions of $Cu(s)$ in H_2SO_4. This last reaction is unusual, because most metals react with solutions of H_2SO_4 to form hydrogen gas and a metal sulfate.
Question:
When sulfuric acid reacts with copper, how does the oxidation number of the sulfur change?

(A) From $+4$ to $+6$

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(B) From +6 to $+4$

(C) From $+6$ to $+8$

(D) From $+8$ to $+6$

Q.13

Paragraph:

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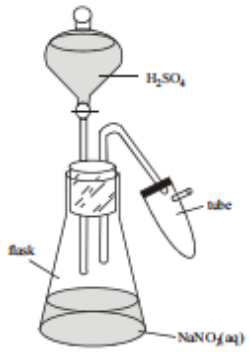
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Question:

The apparatus shown below can be used to prepare HNO_3 (boiling point $= 86^\circ C$)



The yield of HNO_3 collected in the tube can be maximized by maintaining the temperatures of the flask and tube, respectively, at -

(A) $0^\circ C$ and $100^\circ C$

(B) $100^\circ C$ and $0^\circ C$

(C) $350^\circ C$ and $150^\circ C$

(D) $350^\circ C$ and $100^\circ C$

Q.14

Paragraph:

Several features of sulfuric acid are given below.

Preparation of Sulfuric Acid: Sulfuric acid is commonly prepared by the combustion of elemental sulfur to sulfur dioxide, followed by the catalytic oxidation of sulfur dioxide to sulfur trioxide. Sulfur trioxide is then absorbed into a

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98% aqueous solution of H_2SO_4, and water is added to maintain a 98% concentration. SO_3 reacts with the water in the aqueous solution according to Reaction 1. $SO_3(\mathrm{~g})+H_2O(\ell)=H_2SO_4(\ell)$
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Formation of SO_2 : Sulfuric acid forms SO_2 gas when it reacts with several compounds. For example, I_2 and SO_2 are formed when I^{-}reacts with concentrated $H_2SO_4; Br_2$ and SO_2 are formed when Br^{-}reacts with concentrated $H_2SO_4.Cu^{+}$and SO_2 are formed in hot solutions of $Cu(s)$ in H_2SO_4. This last reaction is unusual, because most metals react with solutions of H_2SO_4 to form hydrogen gas and a metal sulfate.

Question:

Which of the following is the balanced equation describing the combustion of elemental sulfur?

(A) $2H_2\ S+3O_2 \rightarrow 2SO_2+2H_2O$

(B) $H_2\ S+2O_2 \rightarrow SO_3+H_2O$

(C) $2SO_3 \rightarrow 2\ S+3O_2$

(D) $S+O_2 \rightarrow SO_2$

Q.15

Paragraph

A steel sample is to be analysed for Cr and Mn simultaneously. By suitable treatment Cr is oxidized as $Cr_2O_7^{2-}$ and the Mn to MnO_4^{-}.

$Cr \quad \rightarrow \quad Cr_2O_7^{2-}$

$Mn \quad \rightarrow \quad MnO_4^{-}$

A $10 \mathrm{gm}$ sample of steel is used to produce 250.0 mL of a solution containing $Cr_2O_7^{2-}$ and MnO_4^{-}. A $10 \mathrm{~mL}$ portion of this solution is added to a $BaCl_2$ solution and by proper adjustment of the acidity, the chromium is completely precipitated as $BaCrO_4$; $0.0549 \mathrm{~g}$ is obtained.

$Cr_2O_7^{2-} \xrightarrow{H^{\oplus}} BaCrO_4$

A second $10 \mathrm{~mL}$ portion of this solution requires exactly $15.95 \mathrm{~mL}$ of $0.0750 \mathrm{M}$ standard Fe^{2+} solution for its titration (in acid solution)**Question**

$\%$ of chromium in the steel sample

(A) 1.496

(B) 2.82

(C) 1.96

(D) 5

Q.16

Paragraph

A steel sample is to be analysed for Cr and Mn simultaneously. By suitable treatment Cr is oxidized as $Cr_2O_7^{2-}$ and the Mn to MnO_4^{-}.

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$Cr \rightarrow Cr_2O_7^{2-}$

$Mn \rightarrow MnO_4^-$

A $10gm$ sample of steel is used to produce 250.0 mL of a solution containing $Cr_2O_7^{2-}$ and MnO_4^-. A $10\ mL$ portion of this solution is added to a $BaCl_2$ solution and by proper adjustment of the acidity, the chromium is completely precipitated as $BaCrO_4$; $0.0549\ g$ is obtained.

$Cr_2O_7^{2-} \xrightarrow{H^{\oplus}} BaCrO_4$

A second $10\ mL$ portion of this solution requires exactly $15.95\ mL$ of $0.0750M$ standard Fe^{2+} solution for its titration (in acid solution)**Question**

Equivalent of Fe^{2+} required for reduction of MnO_4^- is

(A) 5.44×10^{-4}

(B) 0.544×10^{-2}

(C) 1.196×10^{-3}

(D) 11.96×10^{-4}

Q.17

Paragraph

A steel sample is to be analysed for Cr and Mn simultaneously. By suitable treatment Cr is oxidized as $Cr_2O_7^{2-}$ and the Mn to MnO_4^-.

$Cr \rightarrow Cr_2O_7^{2-}$

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A $10gm$ sample of steel is used to produce 250.0 mL of a solution containing $Cr_2O_7^{2-}$ and MnO_4^-. A $10\ mL$ portion of this solution is added to a $BaCl_2$ solution and by proper adjustment of the acidity, the chromium is completely precipitated as $BaCrO_4$; $0.0549\ g$ is obtained.

$Cr_2O_7^{2-} \xrightarrow{H^{\oplus}} BaCrO_4$

A second $10\ mL$ portion of this solution requires exactly $15.95\ mL$ of $0.0750M$ standard Fe^{2+} solution for its titration (in acid solution)**Question**

Amount of $BaCl_2$ required for conversion of $Cr_2O_7^{2-}$ to $BaCrO_4$ in steel sample

(A) 0.045

(B) 0.0549

(C) 1.125

(D) 2.82

Q.18

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Column II gives oxidation number of the underlined elements in Column I. Match them correctly.

Column I	Column II
(A) $\underline{Fe}(CO)_5$	(p) 0
(B) $K_3[\underline{Fe}(CN)_6]$	(q) 1
(C) $(CH_3)_2\underline{S}O$	(r) 2
(D) $\underline{N}OCl$	(s) 3
	(t) 4

(A) A-p; B-s ; C- p ; D-s;

(B) A-p; B-r ; C- s ; D-s;

(C) A-s; B-r ; C-s ; D-s;

(D) A-s; B-p ; C- p ; D-s;

Q.19

A $1.10\ g$ sample of copper ore is dissolved and Cu^{2+} formed are titrated with excess of KI. The liberated iodine requires $12.12ml$ of $0.10\ N Na_2\ S_2O_3$ solution for titration. The $\%$ of copper by mass in sample is...

Q.20

A $1.0\ g$ sample of Fe_2O_3 solid of 55.2% purity is dissolved in acid and reduced by heating the solution with zinc dust. The resultant solution is cooled and made upto $100.0\ mL$. $25.0\ mL$ of this solution requires $17.0\ mL$ of $0.0167\ M$ solution of an oxidant for titration. Calculate the number of moles of electrons taken up by the oxidant in the reaction of the above titration.

Answers & Solutions

Q.1 Answer:

$\frac{M}{61}$

Solution:

$[Fe(CN)_6]^4 \rightarrow Fe^{-3} + CO_2 + NO;$

Inc in $Fe = 3 - 2 = 1$

Inc in $C = 6(+4 - (+2)) = 12$

Inc in $N\cdot = 6(+5 - (-3)) = 48$

Total increment $= 61$

$E = \frac{M}{61}.$

Q.2 Answer:

$NaIO_4$

Solution:

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$8H^{+} + IO_4^{-}(aq) + 7I^{-}(aq) \longrightarrow 4I_2(aq) + 4H_2O(1)$

$x_F = 7 \qquad x_F = 1$

$I_2(aq) + 2Na_2\,S_2O_3(aq) \longrightarrow Na_2\,S_4O_6(aq) + 2NaI(aq)$

$x_F = 2 \qquad x_F = 1$

In the first reaction IO_4^{-} is limiting reagent therefore

$4n_{IO_4^-} = n_{l_2}$

$n_{hypo} = 2n_{l_2}$

$n_{hypo} = n_{1O_4^-} \times 8$

$0.1 \times 1.12 = 8 \times .014$

$.112 = .112$

Q.3 Answer:

$MnO_4^{-} \rightarrow Mn^{+2}$

Solution:

$MnO_4^{-} \longrightarrow Mn^{+2}$

$x + 4 \times -2 = -1$

$x = 7$

Change in oxidation state $= 7 - 2 = 5$

$CrO_4^{-2} \longrightarrow Cr^{+3}$

$x + 4 \times -2 = -2$

$x = 6$

Change in oxidation state $= 6 - 3 = 3$

$\overset{+7}{MnO_4^{-}} \longrightarrow \overset{+4}{MnO_2}$

$x + 2 \times -2 = 0$

$x = 4$

Change in oxidation state $= 7 - 4 = 3$

$^{+6}Cr_2O_7^{-2} \longrightarrow 2Cr^{3+}$

$2x + 7 \times -2 = -2$

$2x = 12$

$x = +6$

Total change in oxidation state $= 2(6 - 3) = 2 \times 3 = 6$

Q.4 Answer:

6

Solution:

Let ferrous oxalate $= x$ g

ferric oxalate $= 2080 - xg$

Moles of FeC_2O_4 = **moles of** $Fe_2\,(C_2O_4)_3$

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$\frac{x}{144} = \frac{2080-x}{376}$

$x = 156$

Moles of FeC_2O_4 = **moles of** $Fe_2(C_2O_4)_3 = 4$

Equivalent of $[FeC_2O_4 + Fe_2(C_2O_4)_3]$ = **Equivalent of** $K_2Cr_2O_7$

$\Rightarrow 4 \times 3 + 4 \times 6 = x \times 6$

$\Rightarrow 36 = 6x$

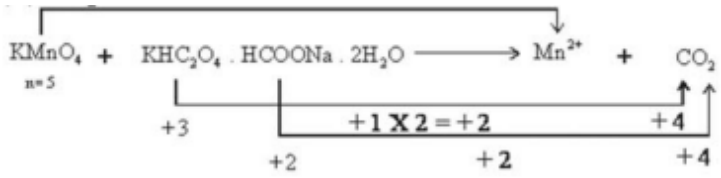
$\Rightarrow x = 6$

Q.5 Answer:

$\frac{4}{5}$

Solution:

Explanation :



$n = +2 + 2 = 4$

n-factor of $KHC_2O_4 \cdot HCOONa2H_2O = 4$

n-factor of $KMnO_4 = 5$

$\therefore$ 5 **moles of** $KHC_2O_4HCOONa$ **reacts with 4 moles of** $KMnO_4$

1 mole of KHC_2O_4**. HCOONa reacts with** $\frac{4}{5}$ **moles of** $KMnO_4$

(b) is correct answer.

Q.6 Answer:

24

Solution:

In the given reaction, Al **is oxidised to o to** $+3$ **oxiation and** F **e is reduced to** $+\frac{8}{3}$ **to 0 .**

Complete balanced equation: $8Al + 3Fe_3O_4 \rightarrow 4Al_2O_3 + 9\ Fe$

In the reaction no.of Al **is 8 . So, number of electrons are transferred is** $8 \times 3 = 24$

Q.7 Answer:

6,6

Solution:

The balanced redox reaction given above can be written as :

$$10FeC_2O_4 + 6KMnO_4 + 24H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 20CO_2 + 6MnSO_4 + 3\ K_2SO_4 + 24H_2O$$

so, the value of $x = 6$ **and** $y = 6$

Q.8 Answer:

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2

Solution:

The reaction are

$$MnO^{-}{}_4 + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O$$

$$A^{+n} + 3H_2O \rightarrow AO_3^- + 6H^+ + (5 - n)e^-$$

Amount of electrons involved in the given amount of $MnO_4^- = 5 \times 1.61 \times 10^{-3}$ mol.

Equating these two we get $5 \times 1.61 \times 10^{-3} = (5 - n)2.68 \times 10^{-3}$

$\therefore n = 2(\text{approx.})$

Q.9 Answer:

$\frac{1}{2}H_2 + \frac{1}{2}I_2 \Leftrightarrow HI$

Solution:

In all the above reaction except (D) there is change in oxidation states of reactant and product atoms, hence they are all redox reactions. In reaction (D) the oxidation states of the atoms of the reactants and products remain unchanged hence, it is not a redox reaction.

Q.10 Answer:

The oxidation number of S in SO_4^{2-} ion is $+6$

Solution:

(a) Let the oxidation number of S **be** x**.**

We know that Ox. no. of $O = -2$

So Ox. no $S + 4$ **(Ox. no.** $O) = -2$

or $x + 4(-2) = -2$

or $x - 8 = -2$

or $x = +8 - 2 = +6$

The oxidation number of S **in** $SO_4{}^{2-}$ **ion is** $+6$**.**

(b) Let the oxidation number of S **be** x **in** HSO_3- **ion.**

We know that, Ox. no. of $H = +1$

Oxidation number of $O = -2$

So Ox. no. $H +$ Ox. no. $S + 3($Ox. no. $O) = -1$

$+1 + x + 3(-2) = -1$

or $+1 + x - 6 = -1$

or $x - 5 = -1$

or $x = +5 - 1 = +4$

The oxidation number of S **in** HSO_3- **ion is** $+4$**.**

(c) Let oxidation number of Pt be x**.**

We know that Ox. no. of $Cl = -1$

So Ox. no. $Pt + 6$ **(Ox. no.** Cl **)** $= -2$

$x + 6(-1) = -2$

or $x - 6 = -2$

The oxidation number of Pt **in** $[Pt(Cl)_6]^{2-}$ **ion is** $+4$**.**

(d) Let oxidation number of Mn **be** x**.**

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We know that Ox. no. of $O = -2$

So Ox. no. $Mn + 4$ (Ox. no. O) $= -1$

$x + 4(-2) = -1$

or $x \quad - \quad 8 \quad = -1$

or $x = +8 - 1 = +7$

The oxidation number of Mn in $[MnO_4]^-$ ion is $+7$.

Q.11 Answer:

% purity of cyanogen is 86.67%

Solution:

$$(NH_4)_2\,CO_3 \rightarrow NH_2 - \underset{\underset{O}{\|}}{C} - NH_2 + 2H_2O$$

$$\text{Moles of urea} = \frac{11.52}{96} = 0.12$$

$$2KMnO_4 + 5\,(NH_4)_2\,C_2O_4 \rightarrow 2Mn^{2+} + 10CO_2 + NH_4^+$$

$$\text{Moles of } (NH_4)_2\,C_2H_4 = 20 \times 1.6 \times (5/2) \times 10^{-3} = 0.08$$

(i) $(CN)_2 + 4H_2O \rightarrow \underset{a}{(NH_4)_2\,C_2H_4}$

(ii) $(CN)_2 + 2H_2O \rightarrow \underset{b}{NH_2CONH_2}$

$a = 0.08 \quad b = 0.12$

Total moles of $(CN)_2 = 0.20$

Weight of $(CN)_2 = 0.20$

Weight of $(CN)_2 = 0.2 \times 52 = 10.4$

(A) % of purity of $(CN)_2 = \frac{10.4}{12} \times 100 = 86.67\%$

(B) % progress in case (i) $= \frac{0.08}{0.20} \times 100 = 40\%$

(C) % progress in case (ii) $= \frac{0.12}{0.20} \times 100 = 60\%$

(D) In case (i), oxidation state remains same. In case (ii), it is oxidized.

Q.12 Answer:

From +6 to $+4$

Solution:

The passage states that sulfuric acid reacts with $Cu(s)$ to produce Cu^+ and SO_2. Thus, sulfuric acid is converted into sulfur dioxide. The oxidation number of sulfur in H_2SO_4 can be found by assigning oxidation numbers of $+1$ for hydrogen and -2 for oxygen. For the formula H_2SO_4 to be neutral, the sum of the oxidation numbers must be zero. If x is the oxidation number of sulfur in H_2SO_4, then: $2(1) + 4(-2) + x = 0$, and $x = +6$.

Likewise, for SO_2 : $2(-2) + x = 0$, and $x = +4$.

The change in oxidation number is from $+6$ to $+4$, which is answer choice b.

Q.13 Answer:

$100^\circ C$ and $0^\circ C$

Solution:

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The boiling point of HNO_3 is given in the question as $86^\circ C$. Because HNO_3 must boil out of the flask and be trapped in the tube, the temperature of the flask must be above the boiling point of HNO_3 (i.e., $< 86^\circ C$) and the temperature of the tube must be less than the boiling point of HNO_3 (i.e, $< 86^\circ C$). Answer b meets these criteria; the other answers do not.

Q.14 Answer:

$S + O_2 \rightarrow SO_2$

Solution:

The combustion of elemental sulfur involves a reaction between oxygen (O_2) and sulfur (S). [Note: Though sulfur exists as S8 molecules, its reactions are normally written in terms of its empirical formula S.] Only Answer d shows such a reaction. Thus, answer choice d is the best answer.

Q.15 Answer:

5

Solution:

for $10ml$

mole of $BaCrO_4 = \frac{0.0549}{253} = 2.17 \times 10^{-4}$

$Cr_2O_7^{2-} \longrightarrow BaCrO_4$

By POAC at $Cr \Rightarrow$ **mole of** $Cr_2O_7^{2-} = \frac{2.17 \times 10^{-4}}{2}$

$= 1.085 \times 10^{-4}$

$\Rightarrow$ **mole of** Cr **present in** $10ml = 1.085 \times 10^{-4} \times 2$

$\Rightarrow$ **mole of** Cr **present in original sample**

(250ml or 10 **g)**

$$= 2 \times 1.085 \times 10^{-4} \times \frac{250}{10} = 5.425 \times 10^{-3} \text{ mole}$$

$$= 5.425 \times 52 \times 10^{-3} \text{ g} = 0.2821 \text{ g}$$

$\%$ **of** $Cr = \frac{0.2821}{10} \times 100 = 2.821\%$

for $10ml$

*** m.eq. of** Fe^{2+} = **m.eq. of** $Cr_2O_7^{2-} + m$**.eq. of** MnO_4^-

$5.95 \times 0.075 \times 1 = 1.085 \times 10^{-4} \times 6 \times 10^{+3}$ + m.eq. of MnO_4^-

n factor for m eq.

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$\Rightarrow$ m.eq. of MnO_4^- $= 1.19625 - 0.651 = 0.54525$

eq. of Fe^{2+} required $= 0.54525 \times 10^{-3} = 5.45 \times 10^{-4}$

mole of $BaCrO_4$ produced $= 2.17 \times 10^{-4}$ for 10ml portion

mole of $BaCrO_4$ for 250ml or 10 g steel sample

$= 2.17 \times 10^{-4} \times \frac{250}{10}$

$= 5.425 \times 10^{-3}$ mole

$\Rightarrow$ mole of BaC_2 required $= 5.425 \times 10^{-3}$ mole

$\Rightarrow$ mass of BaC_2 required $= 5.425 \times 10^{-3} \times 208$

$= 1.128$ g

Q.16 Answer:

11.96×10^{-4}

Solution:

for 10ml

mole of $BaCrO_4 = \frac{0.0549}{253} = 2.17 \times 10^{-4}$

$Cr_2O_7^{2-} \longrightarrow BaCrO_4$

By POAC at Cr $\Rightarrow$ **mole of** $Cr_2O_7^{2-} = \frac{2.17 \times 10^{-4}}{2}$

$= 1.085 \times 10^{-4}$

$\Rightarrow$ **mole of** Cr **present in** 10ml $= 1.085 \times 10^{-4} \times 2$

$\Rightarrow$ **mole of** Cr **present in original sample**

(250ml or 10 g**)**

$= 2 \times 1.085 \times 10^{-4} \times \frac{250}{10} = 5.425 \times 10^{-3}$ mole

$= 5.425 \times 52 \times 10^{-3}$ g $= 0.2821$ g

% **of** Cr $= \frac{0.2821}{10} \times 100 = 2.821\%$

for 10ml

* **m.eq. of** Fe^{2+} = **m.eq. of** $Cr_2O_7^{2-} + m$**.eq. of** MnO_4^-

$5.95 \times 0.075 \times 1 = 1.085 \times 10^{-4} \times 6 \times 10^{+3}$ + m.eq. of MnO_4^-

n factor for m eq.

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$\Rightarrow$ m.eq. of MnO_4^- $= 1.19625 - 0.651 = 0.54525$

eq. of Fe^{2+} required $= 0.54525 \times 10^{-3} = 5.45 \times 10^{-4}$

mole of $BaCrO_4$ produced $= 2.17 \times 10^{-4}$ for 10ml portion

mole of $BaCrO_4$ for 250ml or 10 g steel sample

$= 2.17 \times 10^{-4} \times \frac{250}{10}$

$= 5.425 \times 10^{-3}$ mole

$\Rightarrow$ mole of BaC_2 required $= 5.425 \times 10^{-3}$ mole

$\Rightarrow$ mass of BaC_2 required $= 5.425 \times 10^{-3} \times 208$

$= 1.128$ g

Q.17 Answer:

2.82

Solution:

for 10ml

mole of $BaCrO_4 = \frac{0.0549}{253} = 2.17 \times 10^{-4}$

$Cr_2O_7^{2-} \longrightarrow BaCrO_4$

By POAC at Cr $\Rightarrow$ **mole of** $Cr_2O_7^{2-} = \frac{2.17 \times 10^{-4}}{2}$

$= 1.085 \times 10^{-4}$

$\Rightarrow$ **mole of** Cr **present in** 10ml $= 1.085 \times 10^{-4} \times 2$

$\Rightarrow$ **mole of** Cr **present in original sample**

(250ml or 10 g**)**

$= 2 \times 1.085 \times 10^{-4} \times \frac{250}{10} = 5.425 \times 10^{-3}$ mole

$= 5.425 \times 52 \times 10^{-3}$ g $= 0.2821$ g

% **of** Cr $= \frac{0.2821}{10} \times 100 = 2.821\%$

for 10ml

* **m.eq. of** Fe^{2+} = **m.eq. of** $Cr_2O_7^{2-}$ + *m*.**eq. of** MnO_4^-

$5.95 \times 0.075 \times 1 = 1.085 \times 10^{-4} \times 6 \times 10^{+3}$ + m.eq. of MnO_4^-

n factor for m eq.

$\Rightarrow$ m.eq. of MnO_4^- $= 1.19625 - 0.651 = 0.54525$

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$\Rightarrow$ mole of BaC_2 required $= 5.425 \times 10^{-3}$ mole

$\Rightarrow$ mass of BaC_2 required $= 5.425 \times 10^{-3} \times 208$

$= 1.128$ g

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Q.18 Answer:

A-p; B-s ; C- p ; D-s;

Solution:

((A)- p,(B)-s,(C)-p,(D)-s)

(a) The O.N. contribution of $CO = 0$. **Let the** $O.N.$ **of** $Fe = x$ $x + 5 \times 0 = 0$ **or** $x = 0$

the O.N. of Fe in Fe $(CO)_5 = 0$

(b) The charge of complex $= -3$

The O.N. contribution of $CN = -1$

Let the O.N. of $Fe = x$

$x + (-1) \times 6 = -3$ **or** $x = 6 - 3 = 3$

The O.N. of Fe **in** $K_3\left[Fe(CN)_6\right] = +3$.

(c) The O.N. contribution of $CH_3 = +1$ **(contribution of O.N. by each alkyl group** $= 1$**),**

Let the $O.N.$ **of** $S = x$

$1 \times 2 + x - 2 = 0$ or $x = 0$

The $O.N.$ **of** S **in** $(CH_3)_2\, SO = 0$.

(d) Let the oxidation number of $N = x$

$x - 2 - 1 = 0$ or $x = 3$

The O.N. of N **in** $NOCl = +3$.

Q.19 Answer:

—

Solution:

$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$

Equivalent mass of Cu^{2+} **= molar mass**

Equivalent mass of $S_2O_3^{2-}$ **= molar mass**

Thus, millimoles of Cu^{2+} **= millimoles of** $S_2O_3^{2-}$

$= 12.12 \times 0.1$

$= 1.212$ millimol

$= 1.212 \times 10^{-3}$ mol

$= 1.212 \times 10^{-3} \times 63.5$ g pure Cu

$= 0.076962$ g in 1.1 g Cu − ore

Thus $\%$ **of copper** $= \frac{0.076962}{1.1} \times 100 = 7\%$

Q.20 Answer:

—

Solution:

$Fe_2O_3 = \frac{1 \times 55.2}{100} = 0.552$ g

Aften Zn **dust treatment**

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$Fe^{3+} \longrightarrow Fe^{2+}$

$\downarrow \qquad \downarrow$

$\left.\begin{matrix} \frac{0.552}{160} \times 2\,\text{mole} & \frac{0.552}{160} \times 2\,\text{mole} \end{matrix}\right\}$ in 100 ml

$\Rightarrow$ **In** 25ml

$Fe^{2+} = \frac{0.552 \times 2}{160 \times 4}$ mole ; $(nf = 1)$

Now eq. of Fe^{2+} $=$ **Eq. of oxidant**

$$\Rightarrow \frac{0.552 \times 2}{160 \times 4} \times 1 = \frac{17 \times 0.0167}{1000} \times \text{nf}$$

$$\Rightarrow \quad \text{nf} = 6 \text{ for oxidant}$$

$$\Rightarrow \quad 6 \text{ mole } e^{-}$$

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